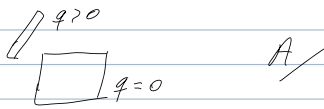


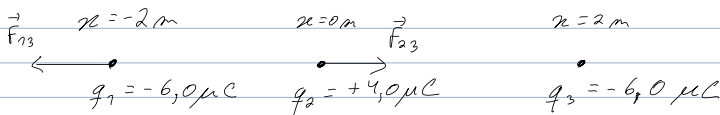
Ersetze multipla

[1]



[2] E/

[3]



$$F_{13} = k \cdot \frac{|q_1| \cdot |q_3|}{d_{13}^2} = 9 \times 10^9 \cdot \frac{|-6,0 \times 10^{-6}| \cdot |-6,0 \times 10^{-6}|}{4^2} = 2,03 \times 10^{-2} \text{ N}$$

$$F_{23} = 9 \times 10^9 \cdot \frac{|+4,0 \times 10^{-6}| \cdot |-6,0 \times 10^{-6}|}{2^2} = 5,40 \times 10^{-2} \text{ N}$$

$$|\vec{F}_{13}| = |-F_{13} + F_{23}| = 3,37 \times 10^{-2} \text{ N}$$

E/

[4] D/

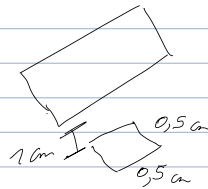
[5] C/

Perguntas

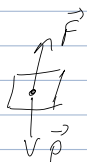
[1]

O papel tem 80 g/m^2

Qual a carga no acetato (q)?



No momento em que o papel é atraído, a força normal é nula mas ainda não existe movimento, logo:



$$F - P = m \cdot a^0 \Leftrightarrow F = P$$

$F \rightarrow$ força do acetato sobre o papel
 $P \rightarrow$ peso do papel

$$m_{\text{papel}} = 80 \times 0,0005^2 = 2,0 \times 10^{-5} \text{ kg} = 2,0 \times 10^{-8} \text{ kg}$$

$$P_{\text{papel}} = m_{\text{papel}} \cdot g = 2,0 \times 10^{-8} \times 9,8 = 1,96 \times 10^{-5} \text{ N}$$

assumindo que as cargas no acetato e no papel são iguais, q :

$$F = k \cdot \frac{|q| \cdot |q|}{d^2} = \frac{9 \times 10^9 \cdot q^2}{0,01^2} = 9 \times 10^{13} \cdot q^2$$

$$F = P \Leftrightarrow 9 \times 10^{13} \cdot q^2 = 1,96 \times 10^{-5} \Leftrightarrow q = \pm 4,67 \times 10^{-10} \text{ C}$$

logo, a ordem de grandeza da carga é de 10^{-10} C

[2]

$$q_1 + q_2 = 10 \mu\text{C}$$

$$F_{12} \Big|_{d=3\text{m}} = 24 \text{ mC} = 9 \times 10^9 \cdot \frac{|q_1| \cdot |q_2|}{3^2} \Leftrightarrow |q_1| \cdot |q_2| = 2,4 \times 10^{-11} \text{ C}^2$$

a) Ambas cargas positivas $\rightarrow q_1 > 0$ e $q_2 > 0 \rightarrow q_1 \cdot q_2 = 2,4 \times 10^{-11} \text{ C}^2$

$$\begin{cases} q_1 = 10\mu - q_2 \\ (10\mu - q_2) \cdot q_2 = 2,4 \times 10^{-11} \end{cases} \Leftrightarrow \begin{cases} \text{---} \\ q_2^2 - 10\mu \cdot q_2 + 2,4 \times 10^{-11} = 0 \end{cases} \Rightarrow$$

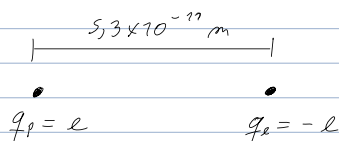
$$\Rightarrow \begin{cases} \text{---} \\ q_2 = 6 \mu\text{C} \vee q_2 = 4 \mu\text{C} \end{cases} \Rightarrow \begin{cases} \text{---} \\ q_1 = 4 \mu\text{C} \vee q_1 = 6 \mu\text{C} \end{cases}$$

le) Uma das cargas positiva e outra negativa $\rightarrow q_1 \cdot q_2 = -2,4 \times 10^{-11} \text{ C}^2$

$$\begin{cases} q_1 = 10 \mu - q_2 \\ (10 \mu - q_2) = -2,4 \times 10^{-11} \end{cases} \Rightarrow \begin{cases} \text{---} \\ q_2^2 - 10 \mu \cdot q_2 - 2,4 \times 10^{-11} = 0 \end{cases} \Rightarrow$$

$$\Rightarrow \begin{cases} \text{---} \\ q_2 = 12 \mu\text{C} \vee q_2 = -2 \mu\text{C} \end{cases} \Rightarrow \begin{cases} q_1 = -2 \mu\text{C} \vee q_1 = 12 \mu\text{C} \end{cases}$$

[3]



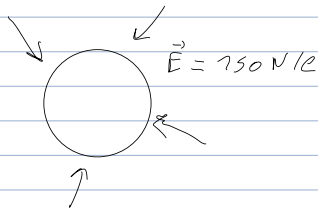
$$e = 1,6 \times 10^{-19} \text{ C}$$

Determinar o módulo do campo elétrico no sítio do elétron.

$$F_{pe} = 9 \times 10^9 \cdot \frac{|1,6 \times 10^{-19}| \cdot |-1,6 \times 10^{-19}|}{(5,3 \times 10^{-11})^2} = 8,20 \times 10^{-8} \text{ N}$$

$$E = \frac{F}{q} = \frac{8,20 \times 10^{-8}}{1,6 \times 10^{-19}} = 5,13 \times 10^{11} \text{ N/C}$$

[4]



$$\begin{aligned} \frac{P_{\text{elétrico}}}{|F_{\text{elétrico}}|} &= \frac{9,109 \times 10^{-31} \times 9,8}{750 \times 1,6 \times 10^{-19}} \quad (F = E \cdot q) \\ &= 3,72 \times 10^{-13} \text{ N/N} \end{aligned}$$

[5]



$$\begin{aligned} q_1 &= 3,2 \text{ mC} \\ q_2 &= 5,1 \text{ mC} \\ q_3 &= 7,4 \text{ mC} \end{aligned}$$

distância entre corpos = 2,65 m

$$F_{21} = 9 \times 10^9 \cdot \frac{|5,7m| \cdot |3,2m|}{(2,65 \times 10^{-2})^2} = 2,09 \times 10^{-4} N$$

$$F_{31} = 9 \times 10^9 \cdot \frac{|7,4m| \cdot |3,2m|}{(2 \times 2,65 \times 10^{-2})^2} = 7,59 \times 10^{-5} N$$

$$F_{13} = F_{31} = 7,59 \times 10^{-5} N$$

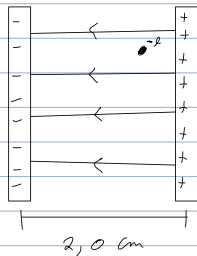
$$F_{23} = 9 \times 10^9 \cdot \frac{|5,7m| \cdot |7,4m|}{(2,65 \times 10^{-2})^2} = 4,84 \times 10^{-4} N$$

Como o sistema está em equilíbrio:

$$T_a = F_{21} + F_{31} = 2,09 \times 10^{-4} + 7,59 \times 10^{-5} = 2,85 \times 10^{-4} N$$

$$T_b = F_{13} + F_{23} = 7,59 \times 10^{-5} + 4,84 \times 10^{-4} = 5,60 \times 10^{-4} N$$

[6]



$$\begin{aligned} a &= \text{const.} \\ v_0 &= 0 \text{ m/s} \\ p_e &\approx 0 \text{ N} \\ \Delta t &= 15 \mu s \\ m_e &= 9,109 \times 10^{-31} \text{ kg} \end{aligned}$$

$$a) |\vec{E}| = ? \rightarrow E = \frac{F_q}{q} \rightarrow F_q = m \cdot a \rightarrow a = \text{const} \rightarrow p(x) = \cancel{v_0}^0 + \cancel{v_0}^0 \cdot t + \frac{1}{2} \cdot a \cdot t^2$$

$$\rightarrow a = \frac{2 \cdot p_f}{t^2} = \frac{2 \times 0,02}{(15 \times 10^{-6})^2} = 1,78 \times 10^8 \text{ m/s}^2$$

$$\rightarrow F_q = 9,109 \times 10^{-31} \times 1,78 \times 10^8 = 1,62 \times 10^{-22} N$$

$$\rightarrow E = \frac{1,62 \times 10^{-22}}{1,6 \times 10^{-19}} = 1,01 \times 10^{-3} \text{ N/C}$$

b)

$$v(t) = \cancel{v_0}^0 + a \cdot t \Leftrightarrow v_f = 1,78 \times 10^8 \times 15 \times 10^{-6} = 2,67 \times 10^3 \text{ m/s}$$

[7]

$x=0$	$x=d$	$x=l$
$\bullet$	$\bullet$	$\bullet$
$q_1 = q$	$q_2 = 2q$	$q_3 = k$

$$\left\{ \begin{array}{l} F_{21} + F_{31} = 0 \Leftrightarrow 9 \times 10^9 \cdot \frac{|2q| \cdot |q|}{d^2} + 9 \times 10^9 \cdot \frac{|k| \cdot |q|}{l^2} = 0 \\ \\ F_{12} + F_{32} = 0 \Leftrightarrow 9 \times 10^9 \cdot \frac{|q| \cdot |2q|}{d^2} + 9 \times 10^9 \cdot \frac{|k| \cdot |2q|}{(l-d)^2} = 0 \end{array} \right.$$

$$\Leftrightarrow \frac{2 \cdot q^2}{d^2} + \frac{k \cdot q}{l^2} = 0 \Leftrightarrow \boxed{\frac{2 \cdot q}{d^2} + \frac{k}{l^2} = 0}$$

$$\Leftrightarrow \frac{2 \cdot q^2}{d^2} + \frac{2 \cdot k \cdot q}{(l-d)^2} = 0 \Leftrightarrow \boxed{\frac{2q}{d^2} + \frac{2 \cdot k}{(l-d)^2} = 0}$$

$$\Leftrightarrow \frac{2 \cdot q}{d^2} + \frac{k}{l^2} = \frac{2q}{d^2} + \frac{2k}{(l-d)^2} \Leftrightarrow k \cdot l^2 + k \cdot d^2 - 2k \cdot l \cdot d = 2 \cdot k \cdot l^2 \Leftrightarrow$$

$$\Leftrightarrow k \cdot l^2 - k \cdot d^2 + 2k \cdot l \cdot d = 0 \Leftrightarrow l^2 + 2 \cdot d \cdot l - d^2 = 0 \Leftrightarrow$$

$$\Leftrightarrow l = 0,474 \cdot d \vee l = -2,47 \cdot d$$

• $l = 0,474 \cdot d$

$$\left\{ \begin{array}{l} \frac{2 \cdot q}{d^2} + \frac{k}{(0,474 \cdot d)^2} = 0 \\ \frac{2 \cdot q}{d^2} + \frac{2k}{(0,474 \cdot d - d)^2} = 0 \end{array} \right. \Leftrightarrow \left\{ \begin{array}{l} \frac{2 \cdot q}{d^2} + \frac{k}{0,224^2 \cdot d^2} = 0 \\ \frac{2 \cdot q}{d^2} + \frac{2k}{0,586^2 \cdot d^2} = 0 \end{array} \right. \Leftrightarrow$$

$$\Leftrightarrow \left\{ \begin{array}{l} k = -0,343 \cdot q \\ k = -0,343 \cdot q \end{array} \right.$$

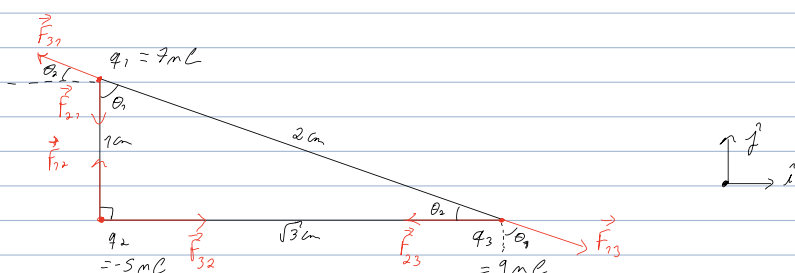
• $l = -2,47 \cdot d$

$$\left\{ \begin{array}{l} \frac{2 \cdot q}{d^2} + \frac{k}{(2,47 \cdot d)^2} = 0 \\ \frac{2 \cdot q}{d^2} + \frac{2k}{(-2,47d - d)^2} = 0 \end{array} \right. \Leftrightarrow \left\{ \begin{array}{l} \frac{2 \cdot q}{d^2} + \frac{k}{2,47^2 \cdot d^2} = 0 \\ \frac{2 \cdot q}{d^2} + \frac{2k}{3,47^2 \cdot d^2} = 0 \end{array} \right. \Leftrightarrow$$

$$\Leftrightarrow \left\{ \begin{array}{l} k = -17,6 \cdot q \\ k = -17,6 \cdot q \end{array} \right.$$

8)

a)



$$F_{21} = 9 \times 10^9 \cdot \frac{|-5 \text{ mC}| \cdot |7 \text{ mC}|}{(0,01)^2} = 3,15 \times 10^{-3} \text{ N}$$

$$F_{31} = 9 \times 10^9 \cdot \frac{|9 \text{ mC}| \cdot |7 \text{ mC}|}{(0,02)^2} = 7,42 \times 10^{-3} \text{ N}$$

$$F_{12} = F_{21}$$

$$F_{32} = 9 \times 10^9 \cdot \frac{|9 \text{ mC}| \cdot |-5 \text{ mC}|}{(\sqrt{3} \times 10^{-2})^2} = 7,35 \times 10^{-3} \text{ N}$$

$$F_{13} = F_{31}$$

$$F_{23} = F_{32}$$

$$\cos(\theta_1) = 1/2 \Rightarrow \theta_1 = 60^\circ$$

$$\cos(\theta_2) = \frac{\sqrt{3}}{2} \Rightarrow \theta_2 = 30^\circ$$

$$\begin{aligned} \vec{F}_1 &= \vec{F}_{21} + \vec{F}_{31} = -3,15 \hat{j} + 7,42 \cdot (\cos(180-30) \cdot \hat{i} + \sin(180-30) \cdot \hat{j}) \\ &= -7,23 \hat{i} - 2,44 \hat{j} \text{ mN} \end{aligned}$$

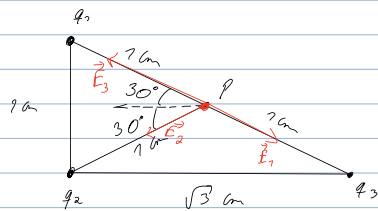
$$\vec{F}_2 = \vec{F}_{12} + \vec{F}_{32} = 7,35 \hat{i} + 3,15 \hat{j} \text{ mN}$$

$$\begin{aligned} \vec{F}_3 &= \vec{F}_{13} + \vec{F}_{23} = 7,42 \cdot (\cos(-90+60) \hat{i} + \sin(-90+60) \hat{j}) - 7,35 \hat{i} = \\ &= -0,120 \hat{i} - 0,710 \hat{j} \text{ mN} \end{aligned}$$

b)

Cada carga produz um campo elétrico $E_i = k \cdot \frac{|q_i|}{d_i^2}$

Logo, no ponto P, o campo total $\vec{E}_P = \vec{E}_1 + \vec{E}_2 + \vec{E}_3$



$$\vec{E}_1 = 9 \times 10^9 \cdot \frac{|7m|}{(0,01)^2} \cdot (\cos(-30)\hat{i} + \sin(-30)\hat{j})$$

$$= (5,46\hat{i} - 3,75\hat{j}) \times 10^5 \text{ N/C}$$

$$\vec{E}_2 = 9 \times 10^9 \cdot \frac{|5m|}{(0,01)^2} \cdot (\cos(180+30)\hat{i} + \sin(180+30)\hat{j})$$

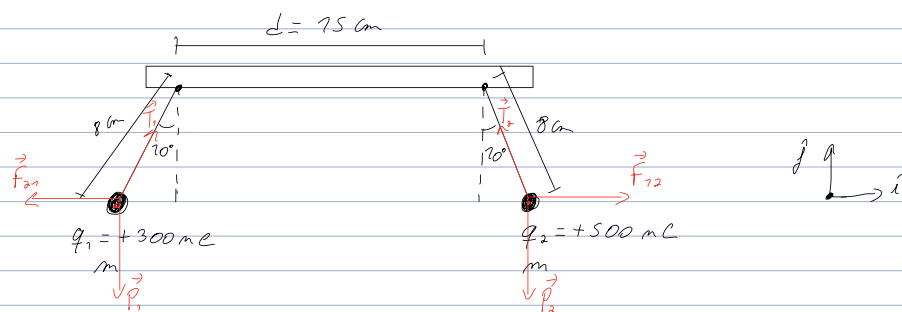
$$= (-3,90\hat{i} - 2,25\hat{j}) \times 10^5 \text{ N/C}$$

$$\vec{E}_3 = 9 \times 10^9 \cdot \frac{|9m|}{(0,01)^2} \cdot (\cos(180-30)\hat{i} + \sin(180-30)\hat{j})$$

$$= (-7,01\hat{i} + 4,05\hat{j}) \times 10^5 \text{ N/C}$$

$$\vec{E}_P = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 = (-5,45\hat{i} - 1,35\hat{j}) \times 10^5 \text{ N/C}$$

9



$$\begin{cases} \vec{P}_1 + \vec{T}_1 + \vec{F}_{21} = m_1 \cdot \cancel{a_1}^0 \\ \vec{P}_2 + \vec{T}_2 + \vec{F}_{12} = m_2 \cdot \cancel{a_2}^0 \\ \vec{P}_1 = \vec{P}_2 = \vec{P} = -9,8m \cdot \hat{j} \end{cases}$$

$$\begin{cases} F_{12} = F_{21} = F = \frac{9 \times 10^9 \cdot |500m| \cdot |300m|}{(0,15 + 2 \cdot 0,08 \cdot \sin(10))^2} = 0,0427 \text{ N} \\ T_1 = T_2 = T \end{cases}$$

$$\begin{cases} -9,8m \cdot \hat{j} + T \cdot (\cos(90-70)\hat{i} + \sin(90-70)\hat{j}) - 0,0427 \cdot \hat{i} = 0 \\ -9,8m \cdot \hat{j} + T \cdot (\cos(90+70)\hat{i} + \sin(90+70)\hat{j}) + 0,0427 \cdot \hat{i} = 0 \end{cases} \Rightarrow$$

$$\begin{cases} (i) \quad 0,774 \cdot T - 0,0427 = 0 \\ (j) \quad 0,985 \cdot T - 9,8 \cdot m = 0 \end{cases} \Rightarrow \begin{cases} T = 0,245 \text{ N} \\ m = 0,0246 \text{ kg} = 24,6 \text{ g} \end{cases}$$
